Sampling complexity of the Random Hierarchy Model under the LS (local score) machine

P* = Theta(|R_L|) = exp(Theta(n)): the LS machine needs training data proportional to the entire grammar.

Criterion (hshunt): a generated grid counts only if it is NOT in the training set AND is fully grammatical.

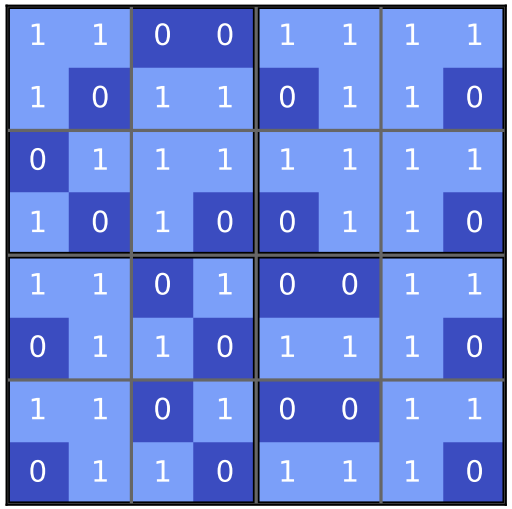
Measured, not derived: the success curve over P was taken for 30 (v, m, L, rule seed) configurations and its peak P* located. Plotted against the exact grammar size |R_L| the peaks collapse: P* spans 63851x, while P*/|R_L| spans 11.9x about a median of 0.197.

This supersedes the 2026-11-02 deck. That one reached exp(Theta(n)) from a ceiling C(k) = f0^U which assumed a window straddling a block boundary constrains nothing. hshunt flagged it, exact enumeration disproved it, and it is withdrawn. The asymptotic answer survives; the route to it is now the data collapse above.

Repo convolutional_diffusion @ 416ea03; rule seeds 1000-1002; alpha = 1e-3; one site revealed at a time (exact ancestral sampling); n_gen = 1024 at L=3, 512-8192 at L=2; jobs 50903850 (exact counts), 50930718, 50935971.

Tokenization: what a "sample" is, and what the window sees

one sample: 8x8 grid of tokens in {0..1}



Tokens. Every site holds one of v = 2 symbols. There is no pixel intensity and no ordering: token 3 is not "between" 2 and 4. A grid is 8x8 = 64 categorical sites.

Grammar. A root symbol expands 3 times. Each symbol is replaced by one of m = 2 synonymous 2x2 "words", drawn once and then fixed. So the grey lines at left bound aligned 2x2 blocks (one level-3 symbol each) and the black lines bound aligned 4x4 blocks (one level-2 symbol each). A grid is grammatical iff some root derives it.

Diffusion. Absorbing-state ("erasure"): each site is independently replaced by a MASK token with probability p(t). Unmasked sites are never corrupted, so exact Bayes denoising is patch matching -- no score approximation enters.

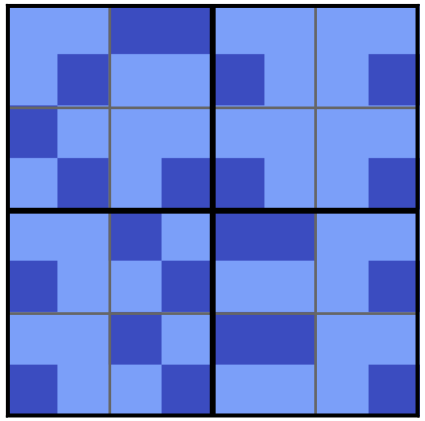
LS machine. The posterior at a masked site is the empirical distribution of the centre tokens of all training k x k patches that agree with the observed sites inside that window, with Laplace smoothing alpha. Sites are revealed one at a time (exact ancestral sampling; revealing in parallel from marginals adds ~n^2/2T independence errors, which alone made 12% of grids ungrammatical).

Success (hshunt's criterion). A generated grid counts only if it is NOT in the training set AND parses completely. Since every training grid is grammatical,

$$\text{success} = \text{grammatical} + \text{novel} - 1$$

Example traces: what the machine actually emits, parsed level by level

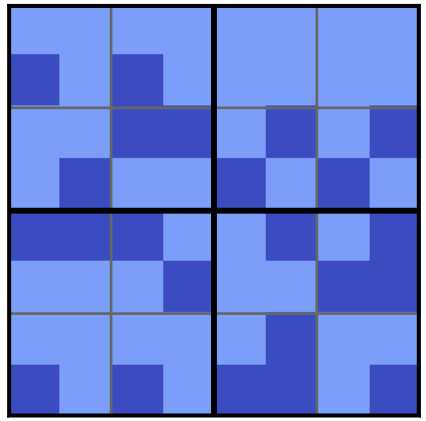
a true RHM sample



```
parse, bottom up
(mean of 64 draws):
  OK leaves      1.00
  OK 2x2 blocks  1.00
  OK 4x4 blocks  1.00
  OK whole grid  1.00

grammatical by construction.
```

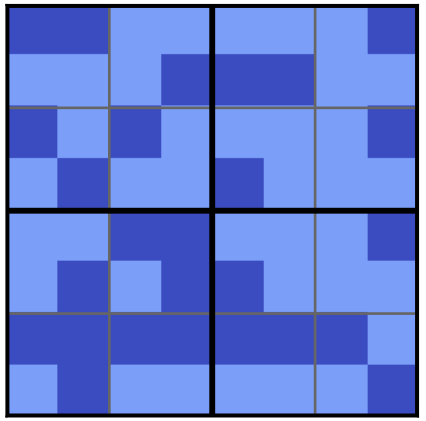
LS machine k=3
(P=20 000)



```
parse, bottom up
(mean of 64 draws):
  OK leaves      1.00
  XX 2x2 blocks  0.55
  XX 4x4 blocks  0.04
  XX whole grid  0.00

reaches 1 site: seals 2x2
blocks, blind to 4x4 and up.
gram 0.00, novel 1.00
```

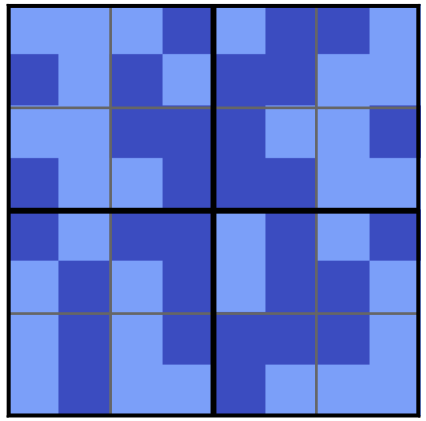
LS machine k=5
(P=20 000)



```
parse, bottom up
(mean of 64 draws):
  OK leaves      1.00
  XX 2x2 blocks  0.52
  XX 4x4 blocks  0.14
  XX whole grid  0.00

reaches 2: still cannot
see the 4x4 rule.
gram 0.00, novel 1.00
```

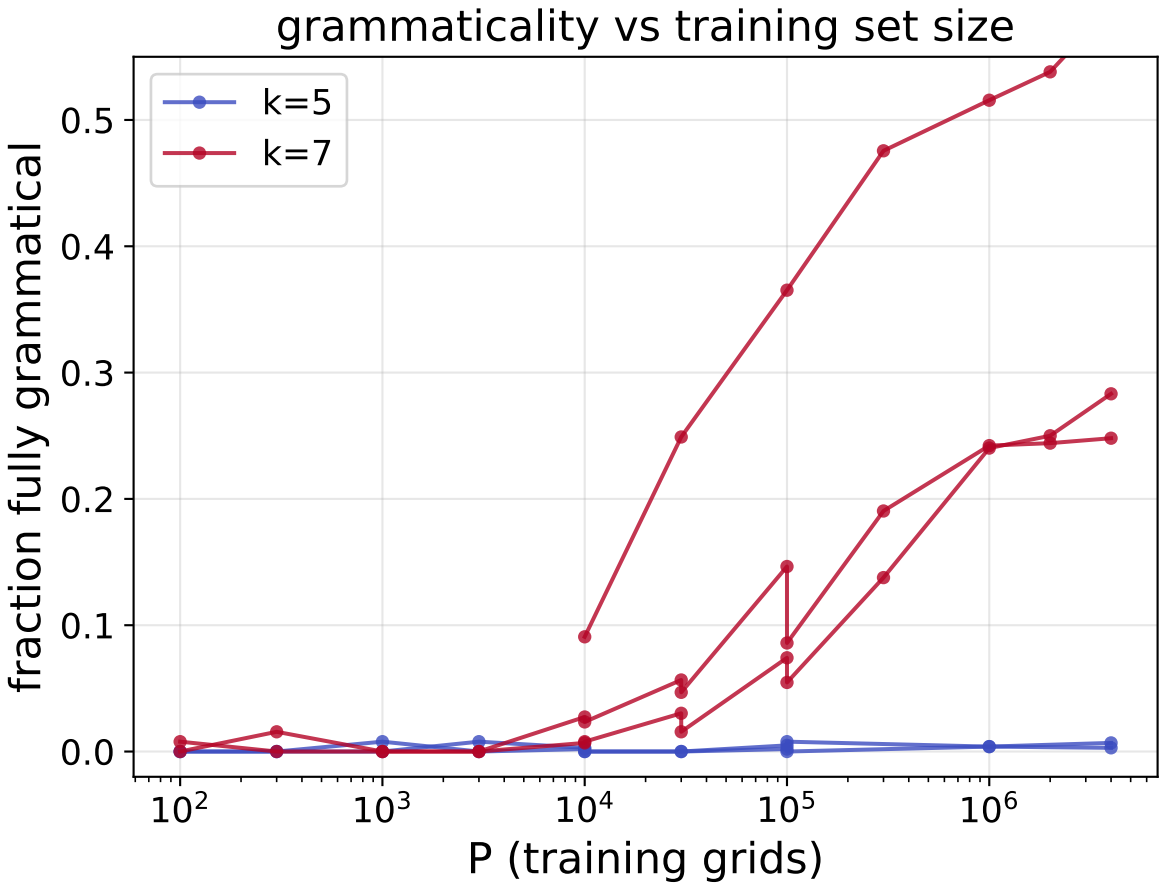
LS machine k=7
(P=20 000)

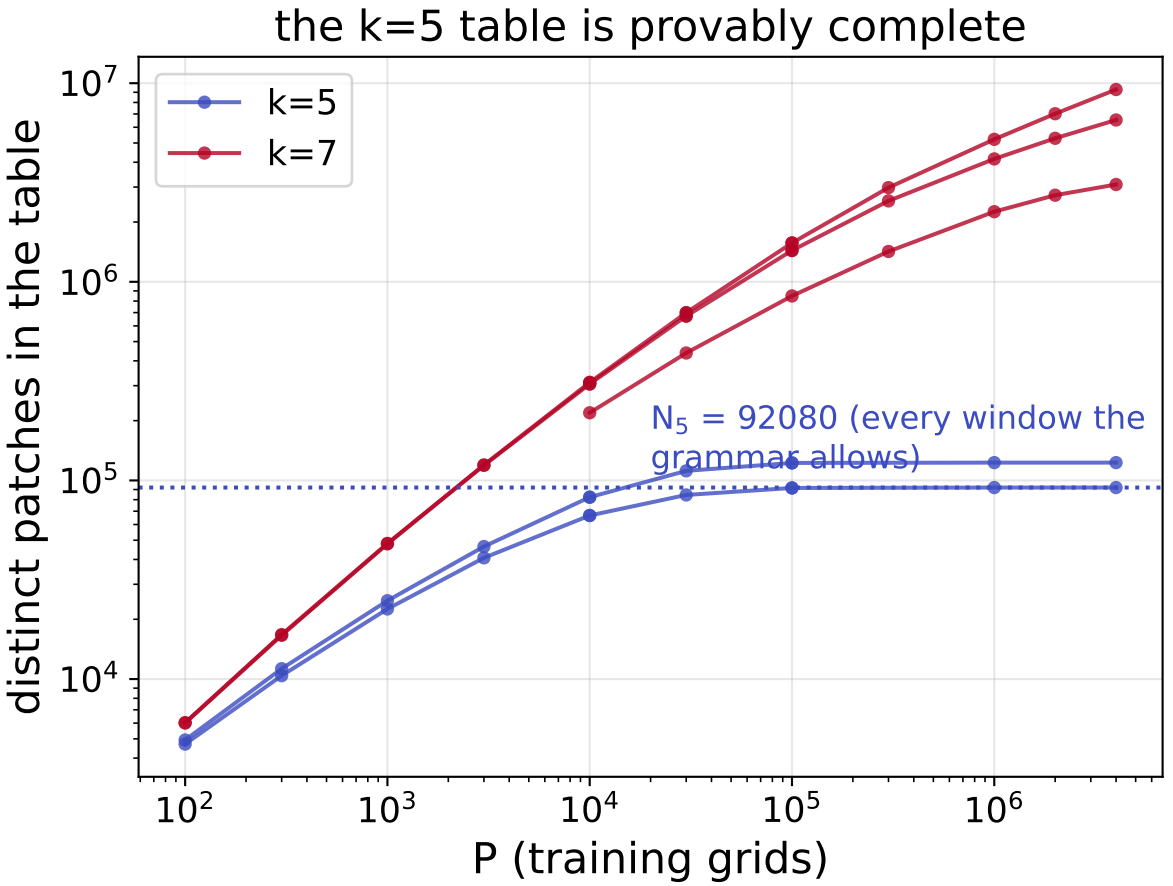


```
parse, bottom up
(mean of 64 draws):
  OK leaves      1.00
  XX 2x2 blocks  0.59
  XX 4x4 blocks  0.24
  XX whole grid  0.03

reaches 3 = U=1 scale:
the first k that can parse
to the root.
gram 0.03, novel 1.00
```

1. A window below the U=1 scale cannot be fixed with data



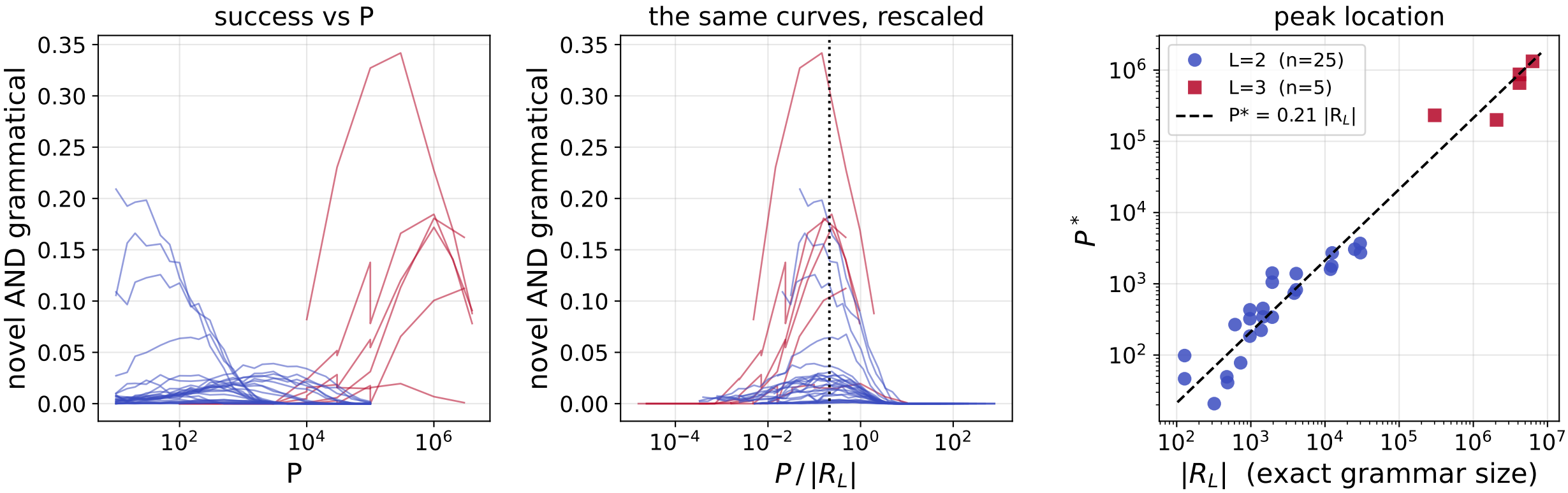


RHM v=2, m=2, s=2, L=3 (8x8 grid, 64 sites), 3 rule seeds, n_gen=1024. A k x k window resolves levels up to j* = floor(log_s((k+1)/2)), so reaching the root needs k >= 2 s^(L-1) - 1 = 7.

At k=5 the table saturates at exactly N_5 = 92080 by P = 1e6 -- it has seen every window the grammar can produce -- and grammaticality never exceeds 0.7% anywhere from P = 1e4 to 4e6, with no trend. At k=7 on the same grammars it reaches 25-52%.

Below the U=1 window, data is not the binding constraint, and no P fixes it. All the sampling-complexity curves that follow are therefore taken at k = U=1.

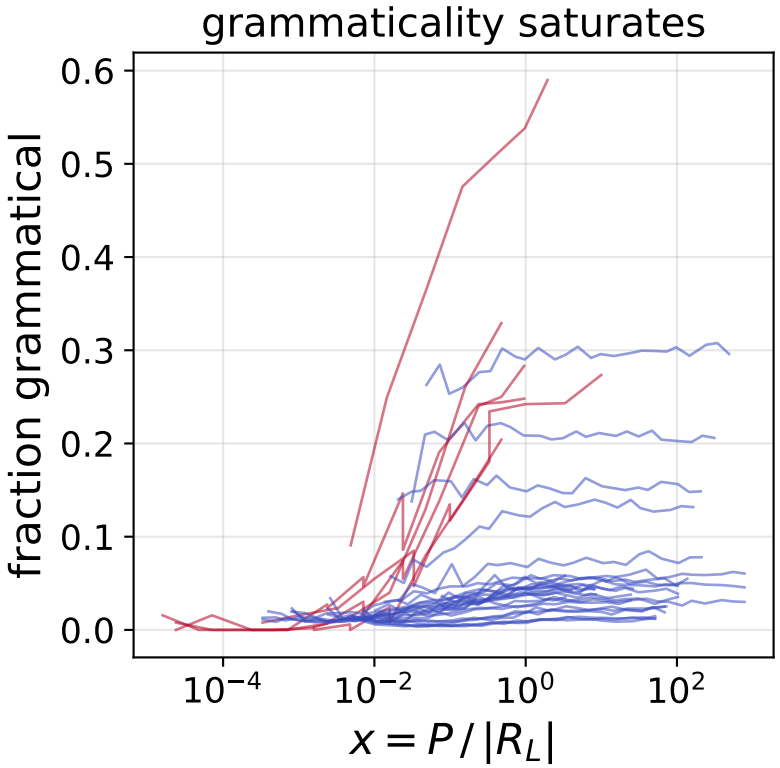
2. Success peaks at P* = 0.21 |R_L|, over five decades of grammar size

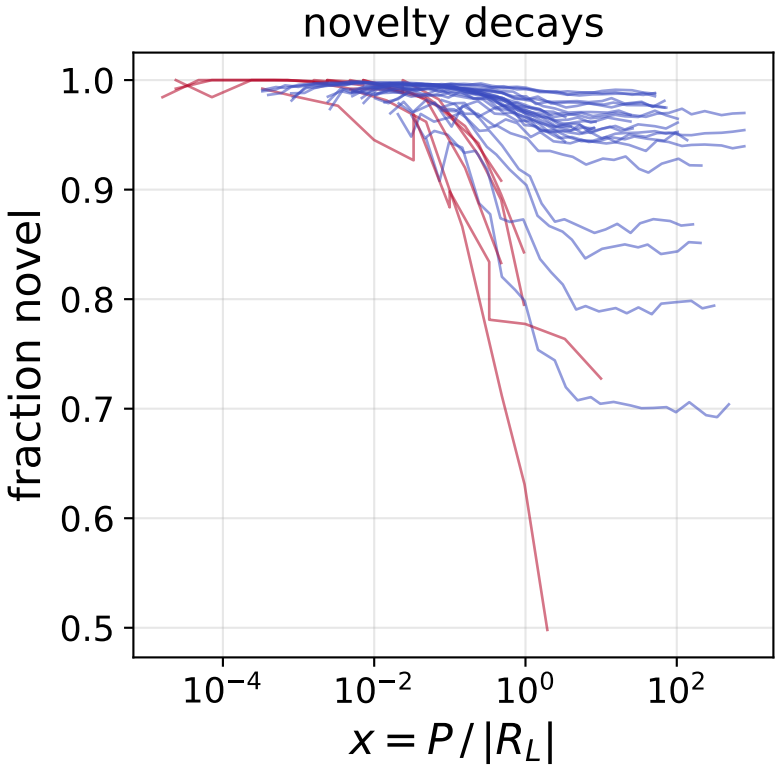


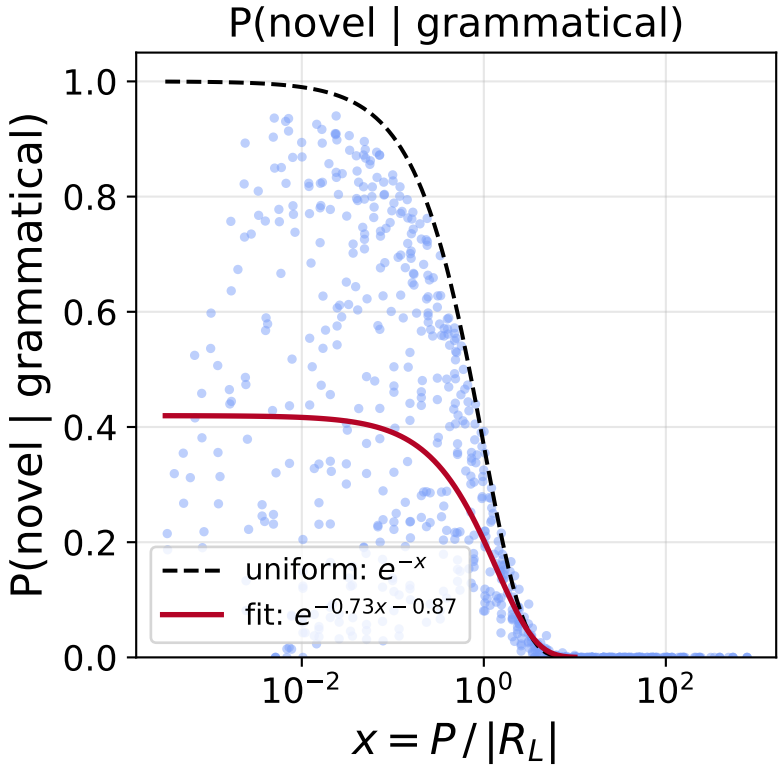
Blue L=2 (16 sites, k=3), red L=3 (64 sites, k=7); each line is one (v, m, rule seed). Rules are drawn with replacement, so independent seeds at the same (v, m) give |R_L| differing by up to 20x -- that is what makes this a collapse rather than a fit. |R_L| is the exact enumerated grammar size, never the nominal v m^M.

```
30 bracketed peaks, v in {2,3,4,6,8}, m in {2,3,4,5,6}, L in {2,3}:
  |R_L| spans 128 - 6 291 456  (49 152 x)      P* spans 21 - 1 326 334  (63 851 x)
  P*/|R_L| spans 0.065 - 0.768  (11.9x), median 0.197, geometric sd 1.95x
  free power law  P* ~ |R_L|^0.965 +- 0.041   -- exponent 1 is 0.8 sd away
```

3. Why a peak exists: a window ceiling that saturates, times novelty that decays







success(x) = grammatical(x) x P(novel | grammatical, x). The first factor rises with data but flattens at a ceiling fixed by the window, not by P -- at L=3, k=7 that ceiling is 0.25-0.52 depending on the rule draw. The second falls. Their product peaks, and because both are functions of x = P/|R_L| alone, the peak sits at a fixed x.

The machine is NOT sampling uniformly from the grammar. Uniform sampling would give P(novel | grammatical) = (1 - 1/|R_L|)^P ~ e^-x (black dashed). Measured: e^(-0.73 x - 0.87) over 488 cells.

The intercept says only 42% of grammatical outputs are novel even as x -> 0: when the LS machine does get the grammar right, it is disproportionately reproducing a training grid. That memorization drag is a second, P-independent penalty, and it is why peak success stays low (median 0.022) even where it is best.

4. The scaling law

$$P^* = \Theta(|R_L|) = \Theta\left(v\, m^{(n-1)/(s^2-1)}\right) = \exp(\Theta(n)), \qquad n = s^{2L} \text{ sites}$$

What was measured

For each (v, m, s, L, rule seed): the full success curve over P at the U=1 window k = 2 s^(L-1) - 1, and the P where it peaks.

Success is hshunt's criterion: NOT in the training set AND parses under every grammar rule, by exact bottom-up parsing.

|R_L| is the exact enumerated grammar size per rule seed, not the nominal v m^M. Rules are drawn with replacement and collide, which is what gives the 20x spread in |R_L| at fixed (v, m) that turns this from a fit into a collapse.

What it says

Two requirements fight. Grammaticality needs the patch table filled, and at the U=1 window that table has N_k = Theta(|R_L|) distinct entries (measured exactly: N_3/|R_2| = 4.3-6.0, N_7/|R_3| = 1.6-6.1). Novelty needs the training set to be a small part of the grammar. Both are functions of x = P/|R_L|, so the optimum sits at a fixed x and P* is linear in the number of grids the grammar can produce -- the LS machine needs data as if the grammar were a flat list, not a hierarchy.

Honest limits

Peak success is low: median 0.022 over 30 curves, best 0.342. The law locates the optimum; it does not claim it is good.

Two depths only (L=2, 16 sites; L=3, 64 sites). L=4 needs k=15, whose table is out of reach, so the exponential in n rests on those two depths plus the exact N_k counts.

The exponent is 0.965 +- 0.041, consistent with 1, but the residual scatter is 1.95x and the constant 0.21 is not sharp.